Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	01 November 2026
Team ID	
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analysis
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Python, HTML, CSS, JavaScript (Tableau calculated-field language for logic)	Python/Flask serves the web page; HTML and CSS style it; Tableau calculated fields create Age Group, Bmi Category and Exercise Label.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	CSV file (smart_healthcare_dataset.csv) stored on Google Drive	The dataset is a single 5,000-row table, so a CSV file is simple and sufficient.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Tableau Data Connection (text file)	Loads the CSV into Tableau Desktop and keeps all 5,000 records.

4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Tableau Desktop (calculated fields, filters, aggregations)	Cleans and groups data and calculates counts, averages and disease rates.
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau Desktop and Tableau Public (KPIs, bar, grouped bar, donut, pie, heat map, dashboard, story)	Interactive charts, dashboard and story that are easy to publish and share.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public and Flask web app (HTML/CSS)	Users interact with filters in the embedded dashboard inside a simple web page.